

ASIM KUMER DEY

Texas Tech University,
Department of Mathematics and Statistics,
Lubbock, TX 79409, United States

Email: a.dey@ttu.edu
[Google Scholar Profile](#), [ORCID](#)

EDUCATION

- 2019 Ph.D., Statistics, The University of Texas at Dallas (UT Dallas), TX.
Advisor: [Yulia R. Gel](#)
- 2015 M.S., Mathematics (Statistics), Lamar University, TX.
- 2010/2009 M.S./B.Sc., Applied Statistics, University of Dhaka, Bangladesh.

EMPLOYMENT/RESEARCH EXPERIENCE

- 08/2024 – Assistant Professor (Tenure Track), Department of Mathematics and Statistics, Texas Tech University.
- 08/2021 – 7/2024 Assistant Professor (Tenure Track), Department of Mathematical Sciences, The University of Texas at El Paso.
- 09/2019 – 07/2021 Visiting Research Collaborator ([H. Vincent Poor](#)'s group), Department of Electrical and Computer Engineering, Princeton University.
- 09/2019 – 07/2021 Postdoctoral Research Associate, The University of Texas at Dallas.
Mentor: Yulia R. Gel
- 08/2017 – 08/2019 NSF Graduate Research Assistant, UT Dallas, TX.
- 06/2018 – 08/2018 NSF-DOE Intern, Pacific Northwest National Laboratory ([PNNL](#)), WA.
- 10/2017 – 12/2017 Visiting Graduate Research Fellow, Statistical and Applied Mathematical Sciences Institute (SAMSI), NC.
- 10/2011 – 07/2013 Software Engineer, IMS Health (now [IQVIA](#)), Dhaka, Bangladesh.

RESEARCH INTERESTS

Statistical methods for complex networks, topological and geometric data analysis, causal inference, Bayesian modeling, deep learning, and extreme value theory, with applications to critical infrastructure networks, medical image analysis, brain networks, epidemiology, climatology, and finance.

AWARDS AND HONORS

- 2021 Summer Institute (Statistics and Modeling in Infectious Diseases) Scholarship, Department of Biostatistics, University of Washington, WA.
- 2020 Best Paper Award, American Statistical Association (ASA) Section for Statistics in Defense and National Security (SDNS).
- 2019 Best Paper Award, ASA Section for Statistics in Defense and National Security (SDNS).
- 2018 The 2nd place in Graduate Student Oral Presentation, 6th Annual Texas STEM Conference, Lamar University, TX.
- 2018 PhD Research Small Grant, The University of Texas at Dallas, TX.

- 2018 Summer Institute (Statistical Genetics) Scholarship,
Department of Biostatistics, University of Washington, WA.
- 2017 Graduate Research Fellowship, SAMSI, NC.
- 2015 The 3rd Place in Graduate Student Poster Session, Conference of Texas Statisticians (COTS),
The University of Texas at Austin, TX.

TRAVEL AWARD

- 2025 Travel Award for the Matroids, Rigidity, and Algebraic Statistics Workshop,
Institute for Computational and Experimental Research in Mathematics (ICERM),
Brown University, RI.
- 2024 Travel Award for the 2024 IMAG/MSM Multiscale Modeling Consortium Meeting:
Setting up TEAMS for Biomedical Digital Twins, NIH Conference Center, MD.
- 2023 Travel Award for the Randomness in Topology and its Applications Workshop,
The Institute for Mathematical and Statistical Innovation (IMSI), IL.
- 2019 Travel Award for the Foundations for Blockchain Data Analytics Workshop, SAMSI, NC.
- 2019 Travel Award for the NSF PIs Workshop for the Algorithms for Modern Power
Systems (AMPS) Program, The George Washington University, Washington, DC.
- 2019 Travel Award for the Conference on Geometric Data Analysis, University of Chicago, IL.
- 2019 ASA Travel Award for the 2019 Joint Statistical Meetings (JSM), Denver, CO.
- 2018 Travel Award for the NSF PIs Workshop for the Algorithms for Modern Power
Systems (AMPS) Program, American University, Washington, DC.
- 2018 Travel Award for the Mathematical and Statistical Methods for Climate and
the Earth System (CLIM) Transition Workshop, SAMSI, NC.
- 2016 Travel Award for the Seminar on Stochastic Processes 2016,
University of Maryland College Park, MD.
- 2015 Travel Award for the UNCG Regional Mathematics and Statistics Conference,
The University of North Carolina at Greensboro, NC.

EXTERNAL FUNDING

- 2020–2021 “RAPID: Collaborative Research: Operational COVID-19 Forecasting
with Multi-Source Information”, NSF DMS 2027793,
Role: Senior Personnel; PI: Yulia R. Gel (University of Texas at Dallas)
- 2020-2021 “UT Dallas NRUF Grant: Market Sensing with Cryptocurrency Chainlets:
Can We Learn More on the Traditional Economy from Non-Traditional Blockchain
Data Sources”, Role: Senior Personnel; PI: Irina Panovska (University of Texas at Dallas).
- 2024-2026 “Subcontract: Open Climate Workbench to support efficient and innovative analysis of
NASA’s high-resolution observations and modeling datasets (21-AIST21_2-0020)”,
NASA JPL 210831, Role: PI (Texas Tech University), \$43,500.
- 2024-2027 “Collaborative research: FDT-BioTech: Aspects of Digital Twin Studies for
Neuroimages”, NSF DMS 2453756, Role: PI (Texas Tech University), \$26,965.
- 2026-2027 “Bayesian Spatio-Temporal Modeling of Extreme Snowfall
in Texas Using Extreme Value Neural Network (EVNN)”, Casualty Actuarial Society
(CAS), USA, Role: PI (Texas Tech University), \$15,000.
- 2026-2029 “Integrating Topological Methods and Statistical Learning for Multiscale
and Multimodal Biomedical Data ”, NSF, USA,
Role: PI (Texas Tech University), \$230,276 (*pending*).
- 2027-2029 “Collaborative Research: AMPS: Scalable Decision-Making for Resilient Power Grid
Operation via Topological and Statistical Learning”, NSF, USA,
Role: PI (Texas Tech University), \$400,147 (*pending*).

- 2026-2029 “Collaborative Research: FDT-BioTech: Leaps in lipid nanoparticle manufacturing using digital-twin enabled design of experiments”, NSF, USA,
Role: CO-PI, PI: Tanushree Roy (Mechanical Engineering, Texas Tech University), \$699,991 (*pending*).
- 2027-2032 “CAREER: Statistical and topological learning of biomedical data across modalities and scales”, NSF, USA, Role: PI (Texas Tech University), \$499,307 (*pending*).

GRANT PROPOSALS SUBMITTED (NOT FUNDED)

- 2025-2028 “Collaborative research: Topological and Geometric Learning for the Analysis and Simulation of Complex Infrastructure Networks”, NSF, USA,
Role: PI (Texas Tech University), \$249,682.
- 2025-2027 “Topological data analysis for substance use and SUD related biomedical data analysis”, NIH, USA, Role: PI (Texas Tech University), \$307,092.
- 2025-2027 “Cancer image learning using topological data analysis”, NIH, USA,
Role: PI (Texas Tech University), \$155,032.
- 2024-2026 “LEAPS-MPS: Topological and Geometric Learning for the Analysis and Simulation of Complex Infrastructure Networks”, NSF, USA,
Role: PI (University of Texas at El Paso), \$250,000.
- 2024-2027 “Collaborative Research: MATH-DT: Graph-embedded Learning-aided Digital Twin for Network Resilience Planning at Multi-Spatio-Temporal Scales”, NSF, USA,
Role: PI (University of Texas at El Paso), \$250,000.
- 2023-2026 “Collaborative Research: Simulation and analysis of power grids based on higher-order network structures”, NSF, USA,
Role: PI (University of Texas at El Paso), \$250,000.
- 2022-2025 “Collaborative Research: Simulation and analysis of power grids based on higher-order network structures”, NSF, USA,
Role: PI (University of Texas at El Paso), \$149,897.
- 2021-2024 “Detecting Dynamic Changes in Multivariate Economic Processes using Topological Data Analysis”, NSF, USA,
Role: Co-PI, PI: Irina Panovska (University of Texas at Dallas), \$149,897.

PUBLICATIONS

Book Chapter

1. Ofori-Boateng, D., Dey, A. K., Gel, Y. R., and Poor, H. V. (2021) “Graph-Theoretic Analysis of Power Grid Resilience”, Advanced Data Analytics for Power Systems, Cambridge University Press, pages 175-194, 2021, <https://doi.org/10.1017/9781108859806.011>

Submitted/In Progress (* The co-authors are my current PhD students)

1. Disanayakage, G. V. W. W.*, and Dey, A. K. (2026+) “Network based Analysis of Disease Spread”, *In Progress*.
2. Mudiyansele, R. T. E. E.*, and Dey, A. K. (2026+) “Bayesian Spatial Causal Inference for County-Level Asthma Prevalence in the United States”, *In Progress*.
3. Dey, A. K., and Panovska, I. (2026+) “A topological data analysis based change point detection in multivariate economic data”, *In Progress*.
4. Dey, A. K., Rajput, A. A., Chen, Y., Gel, Y. R., and Mostafavi, A. (2026+) “Quantifying Resilience of Temporal Mobility Networks Through Topological Features and Motifs”, *In Progress*.

5. Dey, A. K., Goel, J., Chatterjee, S., Gel, Y. R., and Poor, H. V. (2026+) “Topological Validation of Generated Scientific Images”, *Under review*.
6. Rajakaruna, S. N. K.*, Dey, A. K., and Kundu, S. (2026+) “Topology guided conditional generative adversarial network for medical image simulation”, *Under review*.
7. Disanayakage, G. V. W. W.*, and Dey, A. K. (2026+) “Multivariate Functional Inference and Gaussian Mixture Modeling of Joint Temperature-Precipitation Patterns in the U.S. South Central Region”, *Under review*.
8. Lebbe, Z. F. S.*, and Dey, A. K. (2026+) “Efficient Optimal Design for Experiments on Networks under Interference”, *Under review*.
9. Rajakaruna, S. N. K.*, Dey, A. K. and Trindade, A. A. (2026+) “Integrating Persistent Homology and Multilevel Functional Data Analysis for Brain MRI”, *under review*.

Selected Journal Articles

1. Rajakaruna, S. N. K.*, Dey, A. K., Chatterjee, S. (2026) “Enhancing Structural to Functional Brain Network Prediction Using Topological Graph Learning”, *Frontiers in Neuroscience*, published online, <https://www.frontiersin.org/-/10.3389/fnins.2026.1939985/abstract>.
2. Disanayakage, G. V. W. W.*, and Dey, A. K. (2026) “Bayesian Spatial Modeling of Measles Outbreak in Texas, USA”, *Environmental and Ecological Statistics*, 33, 811–834, <https://doi.org/10.1007/s10651-026-00718-5>.
3. Dey, A. K., and Gel, Y. R. (2026) “Integrating Statistical Data Depth and Topological Methods for Anomaly Detection in Weighted Dynamic Networks”, *Technometrics*, 1–14, <https://doi.org/10.1080/00401706.2026.2643214>.
4. Dey, A. K., and Mamun, A. (2026) “Modeling air pollution in Dhaka with Bayesian generalized additive mixed model and quantile regression”, *Environmental and Ecological Statistics*, 33, 653–675, <https://doi.org/10.1007/s10651-026-00720-x>
5. Mustaquim, S. M.*, Dey, A. K., and Mandal, A. (2025) “Understanding How Network Geometry Influences Diffusion Processes in Complex Networks: A Focus on Cryptocurrency Blockchains and Power Grid Networks”, *Journal of the Royal Statistical Society, Series A (Statistics in Society)*, qnaf146, <https://doi.org/10.1093/jrsssa/qnaf146>.
6. Dey, A. K., and Kundu, S. (2025) “Complex network and topological data analysis methods for county level COVID-19 vaccine acceptance analysis in the US”, *Statistics in Medicine*, 44, no. 18-19, e70109, <https://doi.org/10.1002/sim.70109>.
7. Dey, A. K., and Patwary, M.S.A. (2025) “Utilizing Non-stationary Extreme Value Model to Quantify Extreme Rainfall in Two Major Cities in Bangladesh”, *Stochastic Environmental Research and Risk Assessment*, <https://doi.org/10.1007/s00477-025-02969-3>.
8. Dey, A. K., Lyubchich, V., and Gel, Y. R. (2024) “Multivariate Modeling of Precipitation-induced Home Insurance Risks using Data Depth”, *Journal of Agricultural, Biological and Environmental Statistics*, Volume 29, pages 36–55, 2024, <https://doi.org/10.1007/s13253-023-00554-1>.
9. Dey, A. K., Young, S. J., and Gel, Y. R. (2023) “From Delaunay Triangulation to Topological Data Analysis: Generation of More Realistic Synthetic Power Grid Networks”, *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, Volume 186, Issue 3, 2023, Pages 335–354, <https://doi.org/10.1093/jrsssa/qnad066>.
10. Dey, A. K., and Das, K. (2022) “How do mobility restrictions and social distancing during COVID-19 affect the crude oil price?”, *Journal of Statistical Theory and Practice*, 16(22),

2023, <https://doi.org/10.1007/s42519-022-00247-x>.

11. Dey, A. K., Haq, T, Das, K., and Panovska, I. (2022) “Impacts of COVID-19 local spread and Google search trend on the US stock market”, *Physica A: Statistical Mechanics and its Applications*, 589, <https://doi.org/10.1016/j.physa.2021.126423>.
12. Dey, A. K., Tian, Y., and Gel, Y. R. (2022) “Community detection in complex networks: from statistical foundations to data science applications”, *WIREs Computational Statistics*, 14(2), e1566, <https://doi.org/10.1002/wics.1566>.
13. Yuvaraj, M., Dey, A. K., Lyubchich, V., Gel, Y. R., and Poor, H. V. (2021) “Topological Clustering of Multilayer Networks”, *Proceedings of the National Academy of Sciences (PNAS)*, 118(21), <https://doi.org/10.1073/pnas.2019994118>, **(Best Paper Award, ASA SDNS 2020)**.
14. Dey, A. K., Akcora, C. G., Gel, Y. R., and Kantarcioglu M. (2020) “On the Role of Local Blockchain Network Features in Cryptocurrency Price Formation”, *Canadian Journal of Statistics*, 48(3), Pages 561-581, 2020, <https://doi.org/10.1002/cjs.11547>.
15. Dey, A. K., Tiakor, A. and Das, K. (2020) “Determinants of High Crude Oil Price: A Nonstationary Extreme Value Approach”, *Journal of Statistical Theory and Practice*, 14(1),2020, <https://doi.org/10.1007/s42519-019-0070-7>.
16. Dey, A. K., Gel, Y. R., and Poor, H. V. (2019) “What network motifs tell us about resilience and reliability of complex networks”, *Proceedings of the National Academy of Sciences (PNAS)*, 116(39), 19368-19373, <https://doi.org/10.1073/pnas.1819529116>, **(Best Paper Award, ASA SDNS 2019)**.
17. Das, K., and Dey, A. K.(2016) “Quantifying the Risk of Extreme Aviation Accidents.” *Physica A: Statistical Mechanics and its Applications*, 463, pages 55-66, 2016, <https://doi.org/10.1016/j.physa.2016.07.023>.

Peer Reviewed Conference Articles

1. Islambekov, U., Dey, A. K., Gel, Y. R., and Poor, H. V. (2018) “Role of Local Geometry in Robustness of Power Grid Networks”, Proc. 2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP), pages 885-889, 2018.
2. Akcora, C. G., Dey, A. K., Gel, Y. R., and Kantarcioglu M. (2018) “Forecasting Bitcoin Price with Graph Chainlets”, Proc. Advances in Knowledge Discovery and Data Mining (PAKDD), 2018, 765–776, Springer, Cham.
3. Dey, A. K., Gel, Y. R., and Poor, H. V. (2017) “Motif-based analysis of power grid robustness under attacks”, Proc. 2017 IEEE Global Conference on Signal and Information Processing (GlobalSIP), 1015-1019.

Peer Reviewed Workshop Articles

1. Ofori-Boateng, D., Dey, A. K., Gel, Y. R., Li, B., Zhang, J., and Poor, H. V. (2019) “Assessing the resilience of Texas power grid network”, Proc. 2019 IEEE Data Science Workshop, Minneapolis, MN, USA, pp. 280-284.
2. Dey, A. K., Gel, Y. R., and Poor, H. V. (2017) “Intentional Islanding of Power Grids with Data Depth”, Proc. 2017 IEEE 7th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP), 1-5.
3. Dey, A. K., Lyubchich, V., and Gel, Y. R. (2017) “Modeling Weather-induced Home Insurance Risks with Support Vector Machine Regression”, Proc. Workshop on Mining Big Data in

PRESENTATIONS

Invited Research Seminars

1. “Topological Data Analysis and Data Depth for Detecting Anomalies in Temporal Networks”, Texas Tech University, December 5, 2023.
2. “Topological Clustering of Multilayer Networks”, The University of Texas at El Paso, December 9, 2020.
3. “Topological Clustering of Multilayer Networks”, University of New Brunswick (Fredericton), Canada, Oct 27, 2020.
4. “Topological Data Analysis (TDA) for Time Series”, The University of North Carolina at Greensboro. Nov 6, 2020.
5. “Multivariate Risk Modeling for Precipitation-related Home Insurance Claims”, University of Maryland Center for Environmental Science, Oct 30, 2019.
6. “Role of Higher Order Topological Features in Robustness and Functionality of Complex Networks“, Department of Mathematical Sciences, Kent State University, Mar 15, 2019.

Conference Presentations

1. 2026 Joint Statistical Meetings (JSM), Boston, MA, August 2026. “Hypothesis Testing for Comparing Biomedical Image Distributions Using Persistent Homology”.
2. Conference of Texas Statisticians (COTS), The University of Texas at Austin, Austin, TX, April 2026, “A Persistent Homology-based Hypothesis Test for Comparing Images with Application to Digital Twins of Biomedical Images”.
3. 2025 Joint Statistical Meetings (JSM), Nashville, Tennessee, August, 2025. “Topological Clustering of Multilayer Networks” (Invited).
4. Matroids, Rigidity, and Algebraic Statistics Workshop, Institute for Computational and Experimental Research in Mathematics (ICERM), Brown University, RI, March, 2025. “A Persistent Homology-based Hypothesis Test for Comparing Image Distributions with Application to Digital Twins of Biomedical Images”.
5. The 6th SIAM Texas-Louisiana Sectional Meeting (SIAM TX-LA 2023), University of Louisiana at Lafayette, November 2023. “Survival Analysis for Topological Data Analysis Features” (Invited).
6. The International Environmetrics Society (TIES) Regional Meeting, Trent University, Canada, 26 July 2023. “A network analysis approach to evaluating COVID-19 vaccine acceptance in the US” (Invited).
7. American Mathematical Society (AMS) Fall Central Meeting, September 2022. “Topological anomaly detection in a temporal transportation network” (Invited).
8. The 4th Annual Meeting of the SIAM Texas-Louisiana Section, November 2021. “Topological anomaly detection in a temporal transportation network” (Invited).
9. The 4th International Conference on Econometrics and Statistics (EcoSta 2021), June 25, 2021. “Topological Clustering of Multilayer Networks” (Invited).

10. NSF Conference on COVID-19 Modeling, Jan 29, 2021. “Impacts of COVID-19 local spread and Google search trend on the US stock market”.
11. 2020 Joint Statistical Meetings (JSM), virtual (due to COVID-19 pandemic), Aug 3, 2020. “Topological and Geometric Methods for Resilience Analysis of Power Grid Networks” (Invited).
12. The Lloyd Roeling UL Lafayette Mathematics Conference, Lafayette, LA, October 2019 “Determinants of High Crude Oil Price: A Nonstationary Extreme Value Approach”.
13. Joint Algorithms for Threat Detection (ATD) + Algorithms for Modern Power Systems (AMPS) Annual Workshop, The George Washington University, Washington, DC, October 2019. “What Network Motifs Tell Us about Robustness and Reliability of Complex Networks”.
14. Foundations for Blockchain Data Analytics Workshop, Statistical and Applied Mathematical Sciences Institute (SAMSI), Durham, NC, October 2019, “On Role of Local Blockchain Network Features in Cryptocurrency Price Formation”.
15. 2019 Joint Statistical Meetings (JSM), Denver, CO. “What Do Network Motifs Tell Us about Resilience and Reliability of Complex Networks?”.
16. Conference on Geometric Data Analysis, The Stevanovich Center for Financial Mathematics, The University of Chicago, 2019. “What Do Local Geometry Tell Us about Resilience and Reliability of Complex Networks?”.
17. The 6th Annual Texas STEM Conference, 2018, Lamar University, TX. “What Do Network Motifs Tell Us about Robustness and Reliability of Complex Networks?” **(The 2nd place in Graduate Student Oral Presentation)**
18. 2018 NSF PIs Workshop for the Algorithms for Modern Power Systems (AMPS) Program, American University, Washington, DC, October 2018. “Development of synthetic power grid networks”.
19. Conference of Texas Statisticians (COTS), Southern Methodist University, March 2017, Dallas, TX. “Modeling Weather-induced Home Insurance Risks: A Machine Learning Approach”.
20. 17th SIAM International Conference on Data Mining (SDM), April 2017, Houston, TX, USA. “Modeling Weather-induced Home Insurance Risks with Support Vector Machine Regression”.
21. NSF-EDT Kick-Off Meeting, University of Texas at Dallas, October 2016. “Modeling and Forecasting Weather-related House Insurance Claims with Modern Machine Learning Approaches”.
22. 11th Annual UNCG Regional Mathematics and Statistics Conference, October 2015, The University of North Carolina at Greensboro, NC. “Modeling Extreme Aviation Accident using the Generalized Pareto Distribution.”
23. Conference of Texas Statisticians (COTS), March 2015, The University of Texas at Austin, Austin, TX. “Modeling Extreme Hurricane Damage in the United States: Generalized Pareto Model Approach.” **(The 3rd place in Graduate Student Poster Competition)**
24. Conference of Texas Statisticians (COTS), March 2014, University of Texas at Dallas, Dallas, TX. “Quantifying Extreme Hurricane Risk in the United States Gulf Coast”.
25. Education Research Conference, Lamar University, Beaumont, TX, March 2014. “Reading Interest of Middle School Student and Educational Achievement.”

ORGANIZATION OF TECHNICAL MEETINGS

- 2026 Chair of the session “Spatial Methods for Imaging Data”, 2026 Joint Statistical Meetings (JSM), Boston, MA.
- 2022 Co-organizer of the Special Session on “Statistical Methodology and Applications”, American Mathematical Society (AMS) Fall Central Sectional Meeting, El Paso, Texas.
- 2020 Co-organizer of the invited session on “How Statistics and Data Science Help to Quantify Resilience of Power Systems”, Joint Statistical Meetings (JSM), Philadelphia, Pennsylvania.

TEACHING EXPERIENCE

Recent Courses at Texas Tech University

- 1. STAT-5375: Statistical Multivariate Analysis
- 2. STAT 5376-D21: Advanced Statistical Methods: Spatio-Temporal Statistical Methods for Environment and Epidemiology
- 3. STAT 5376-D21: Advanced Statistical Methods: Network Data Analysis
- 4. Math 3342: Mathematical Statistics for Engineers and Scientists
- 5. STAT 5326: Biostatistics

Recent Courses at The University of Texas at El Paso

- 1. STAT 3320: Probability and Statistics
- 2. STAT 5388/6388: Multivariate Data Analysis
- 3. STAT 5385/ STAT 6385: Statistics in Research
- 4. STAT 5381: Mathematical Statistics II
- 5. DS 5381: Mathematical Foundations of Data Science II
- 6. STAT 5474 - Introduction to Data Mining
- 7. STAT 5370 - Special Topics (Statistical Analysis of Network Data)

COMPUTING SKILLS

R, Python, Stan, Julia, MATLAB, SAS, STATA, MS SQL Server.

STUDENT THESES/PROJECTS SUPERVISE

- 1. Rajitha Edirisingha, PhD Thesis, Texas Tech University (Expected graduation - Summer 2029)
- 2. Zuhra Fazroon, PhD Thesis, Texas Tech University (Expected graduation - Summer 2028)
- 3. Shashipraba Rajakaruna, PhD Thesis, Texas Tech University (Expected graduation - Summer 2027)
- 4. Gihani Wickramasinghe, PhD Thesis, Texas Tech University (Expected graduation - Summer 2027)
- 5. S M Mustaqim, PhD Thesis, University of Texas at El Paso (Expected graduation - Summer 2027)
- 6. Nathaniel Sakyi, Master’s Thesis, Spring 2022, The University of Texas at El Paso

PROFESSIONAL SERVICE

Service at Texas Tech University

1. Oversee and update the course Math 3342. (2024 - 2025)
2. PhD Thesis Committee Member, Thilini Weerasekara, “Imprinting and Maternal Effects Detection Based on the Profile Likelihood Method” (2026 -) (Chair: Fangyuan Zhang).
3. Master’s Thesis Committee Member, Md Samiul Islam (2026) (Chair: Leif Ellingson).
4. Master’s Non-Thesis Committee Member, Sadia Afrin. “Dyadic Data Analysis” (2024) (Chair: Alex Trindade).

Service at The University of Texas at El Paso

1. Committee Member - Data Science Program (2023)
2. Committee Member - Data Science Search committee for the Assistant Professor (2021-2022)
3. Committee Member - Statistics Graduate Committee (2021-2024)
4. Committee Member - Undergraduate curriculum committee (2021-2024)

Reviewer

1. Reviewer - NSF Grant Proposal (Program - Statistics, 2024)
2. Reviewer - Climate Informatics 2025, ASA Statistics in Imaging Section student paper competition 2026.
3. Journal Reviewer - Journal of the Royal Statistical Society Series A, Electronic Journal of Statistics, Biometrics, Computational Statistics & Data Analysis, Technometrics, Canadian Journal of Statistics, Physica A: Statistical Mechanics and its Applications, and REVSTAT.

PROFESSIONAL ORGANIZATIONS

- American Statistical Association (ASA)
- Institute of Mathematical Statistics (IMS)
- Mathematical Association of America (MAA)